

DETAILED REVISION CHECKLIST

Medora: An AI-assisted healthcare management and consultation platform

Prepared for SoftwareX journal submission

Manuscript	Medora
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Recommendation	Major revision required before submission
Purpose	Student correction, safety validation, and resubmission planning

Instruction to the student. This checklist requires technical corrections, new evaluation, privacy clarification, software-release work, and manuscript revision. Language editing alone is not sufficient. Do not mark an item complete until the manuscript, deployment, repository, benchmark outputs, privacy documentation, and archived release are consistent.

Prepared: 1 August 2026

1. Executive assessment

Review dimension	Assessment
SoftwareX fit	Strong potential because the software integrates healthcare workflow, prescription OCR, real-time booking, consent control, and bounded AI assistance.
Strongest contribution	Separation of core API, OCR service, provider orchestration, role-aware workflow, and domain-aware prescription normalization.
Primary rejection risk	Absolute privacy and anonymity claims conflict with cloud OCR and are unsupported by redaction, AI-safety, or factuality evaluation.
Evidence gap	OCR dataset composition, component ablation, booking concurrency correctness, AI factuality, consent enforcement, and data governance are insufficiently reported.
Immediate priority	Correct the privacy architecture, rebuild evaluation, archive a fixed release, and remove production-grade or clinically effective wording.

Highest-risk issues

- Raw prescription images may be sent to a cloud OCR provider even though the paper claims that all external AI inputs are anonymized.
- The consent-and-PII guard has no reported bilingual leakage or over-redaction evaluation.
- Medicine-name and dosage accuracy values lack transparent denominators, corpus composition, confidence intervals, and baselines.
- Schema validation and route restriction are presented as sufficient safety controls without adversarial or factuality testing.
- Real-time update latency is treated as evidence of booking consistency without a demonstrated atomic concurrency design.

Current strengths to retain

- The integrated patient, doctor, and administrator workflow addresses a concrete software problem.
- Separating the core API and OCR service is technically sensible for isolation and scaling.
- Medicine-catalog matching and dosage grammar are potentially publishable technical components.
- The manuscript already acknowledges the limited OCR corpus, absence of an independent security audit, and absence of a field pilot.
- The public deployment and benchmark tooling can support a strong reproducibility package after versioned archiving.

Recommended correction sequence

Phase	Focus	Required outcome
1	Privacy and safety	

Phase	Focus	Required outcome
		Separate text, image, and audio paths; remove anonymity guarantees; evaluate redaction and consent; define research/pilot maturity.
2	Core correctness	Implement and test atomic booking, grounded summaries, safe symptom navigation, and secure offline behavior.
3	Evaluation	Rebuild OCR metrics, add ablations, AI-safety tests, concurrency tests, and a reproducible performance protocol.
4	Release and manuscript	Archive the version, add ethics/data statements, revise figures, remove repetition, and complete the final audit.

Severity interpretation

Severity	Meaning
Submission blocker	Must be corrected before submission because it affects privacy, safety, ethics, scientific validity, correctness, or reproducibility.
Major revision	Substantial change required to architecture explanation, evaluation, workflow semantics, or software documentation.
Presentation revision	Improves precision, readability, figure quality, and journal conformity after technical corrections.
Final audit	Evidence that the manuscript, release, supplementary material, and claims are ready for submission.

Recommended title and abstract

Recommended title: *Medora: An open-source, consent-aware platform for healthcare workflow management, prescription digitization, and bounded AI assistance*

Medora is an open-source, mobile-first web platform integrating patient onboarding, doctor verification and search, appointment booking, consultations, prescriptions, reminders, and consent-controlled record sharing. A separate OCR service converts prescription images into reviewable structured fields using region detection, cloud or local OCR, medicine-catalog matching, and dosage-grammar repair. AI-assisted navigation and record summarization are mediated by role, consent, access-control, direct-identifier redaction, and schema-validation checks. Technical evaluation covers OCR field accuracy, API latency, booking correctness and propagation, and frontend performance. Medora is a research and pilot platform rather than a clinically validated or security-certified system, and all extracted or generated clinical content requires human verification.

2. Submission blockers - privacy, safety, and evidence

[] M-C1 Create and archive a fixed software release

Location: *Code metadata and software metadata tables, pp. 2-3.*

Problem: The manuscript lists a GitHub repository and live deployment but no permanent archived code DOI or reproducible capsule.

Why this matters: A SoftwareX paper must refer to the exact version evaluated. A mutable branch and changing cloud deployment cannot serve as the sole scholarly artifact.

Required correction:

- Create a versioned release such as `v1.0.0` and record the exact commit hash.
- Archive the release in Zenodo or another DOI-issuing repository.
- Include benchmark scripts, test fixtures, configuration, model/provider identifiers, raw outputs, and a deterministic mock mode.
- Add `CITATION.cff`, changelog, third-party license notices, and a software citation in the manuscript.
- Verify that the live deployment, screenshots, and reported benchmark correspond to the archived release.

Suggested wording: *All results reported in this article correspond to Medora v1.0.0, commit `<hash>`, archived at `<DOI>`.*

Completion evidence: The DOI resolves to the exact evaluated source and reproducibility package.

[] M-C2 Separate the text-LLM privacy path from the prescription-image OCR path

Location: Abstract; Section 1; Section 2.1; Figures 1-3, pp. 1, 3-6, and 9.

Problem: The manuscript claims that all AI requests are de-identified before leaving the system, yet Azure Document Intelligence is the primary prescription OCR engine. Raw prescription images may contain names, contacts, signatures, clinic details, dates, diagnoses, and identifiers before OCR.

Why this matters: The current architecture description creates an internal contradiction and an unsafe impression of complete anonymization.

Required correction: Document three distinct data paths:

1. **External text-model path:** role and relationship check, consent check, direct-identifier redaction, provider call, schema validation, and sanitized logging.
2. **Prescription-image OCR path:** whether images leave the deployment, provider identity, region, consent, retention settings, local fallback, and result storage.
3. **Voice path:** whether `faster-whisper` runs locally, whether audio leaves the deployment, and whether audio is retained.

Also redraw the architecture to show the external trust boundary and raw-image path explicitly.

Suggested wording: *Text prompts sent to external language-model providers pass through configured consent, access-control, and direct-identifier redaction checks. Prescription images follow a separate OCR path and may be processed by a cloud OCR provider when explicitly enabled; therefore, the system does not claim complete anonymization of all AI inputs.*

Completion evidence: Manuscript, diagram, privacy documentation, configuration, and deployment behavior describe each external data transfer accurately.

[] M-C3 Remove absolute anonymity and non-traceability claims

Location: Abstract; Sections 1, 2.1, 4, and 5, pp. 1, 3, 5, 9, and 12.

Problem: Statements such as *cannot trace a request to a patient*, *leaking nothing identifiable*, *private anonymous path*, and *providers never receive a patient identity* are absolute and unsupported.

Why this matters: Removing direct identifiers does not eliminate quasi-identifiers, rare conditions, free-text details, network metadata, provider logs, or re-identification risk.

Required correction:

- Replace absolute guarantees with measured and bounded claims.
- Define which direct identifiers are detected and removed in Bengali and English.
- State residual risks, assumptions, and provider-side limitations.
- Do not use *anonymous* unless a formal anonymization assessment supports it.
- Correct the same wording in figures, interface descriptions, README, and deployment notices.

Suggested wording: *The guard removes configured direct identifiers and replaces internal account references before external text-model calls. This reduces exposure of identifiable information but does not guarantee anonymity or eliminate all re-identification risk.*

Completion evidence: No absolute privacy guarantee remains anywhere in the article or software documentation.

[] M-C4 Evaluate the PII redaction and consent guard

Location: Section 2.1 and Figure 2, pp. 5-6.

Problem: The consent-and-PII guard is the central safety claim, but no test corpus, detection performance, residual leakage, over-redaction, or consent-enforcement result is reported.

Why this matters: A privacy boundary is credible only when its failure modes and measured limitations are disclosed.

Required correction:

- Create a bilingual synthetic test set containing names, phone numbers, emails, addresses, dates, national/account identifiers, clinician details, mixed scripts, spacing variations, misspellings, and adversarial formatting.
- Report direct-identifier removal recall, precision, false redaction, and residual leakage by category.
- Test missing, expired, revoked, and scope-mismatched consent.
- Test unauthorized clinician access and attempts to request another patient's information.
- Include prompt-injection attempts requesting raw identifiers or system logs.
- Archive the test set and expected outputs.

Suggested wording: *The redaction module was evaluated on a synthetic bilingual identifier corpus. Results are reported by identifier class, including residual leakage and over-redaction, and should not be interpreted as a guarantee of anonymization.*

Completion evidence: A dedicated evaluation table and reproducible test suite support the privacy-boundary claim.

[] M-C5 Rebuild the prescription OCR evaluation with transparent units and dataset composition

Location: Abstract; Section 4; Table 3, pp. 1 and 11-12.

Problem: Approximately 92% medicine-name accuracy and 89% dosage accuracy are reported without the exact number of prescriptions, fields, writers, languages, image conditions, or confidence-threshold coverage.

Why this matters: The percentages cannot be independently interpreted, reproduced, or compared.

Required correction:

- Report numbers of prescriptions, writers/doctors, medicine fields, dosage fields, quantities, durations, complete medication rows, and complete prescriptions.
- Describe Bengali/English distribution, printed/handwritten mix, devices, resolution, blur, rotation, lighting, cropping, and handwriting variability.
- Describe ground-truth preparation, annotators, and agreement.
- Define raw exact match, normalized medicine match, dosage match, and accepted fuzzy match.
- Report 95% confidence intervals, confidence-threshold coverage, review rate, full-line accuracy, and full-prescription exact match.
- Separate OCR recognition from domain normalization so the contribution of each can be measured.

Suggested wording: *Evaluation units are reported separately for raw OCR text, normalized medicine names, dosage schedules, quantities, durations, complete medication rows, and complete prescriptions.*

Completion evidence: Every reported percentage has a transparent denominator and archived field-level output.

[] M-C6 Add OCR baselines and component ablation

Location: Sections 1, 2.2.4, and 4, pp. 3, 7, and 9-12.

Problem: The paper claims that region detection, medicine-catalog matching, dosage grammar, and Bengali numeral normalization produce the observed accuracy, but no experiment isolates their contribution.

Why this matters: The central technical contribution must be demonstrated rather than asserted.

Required correction: Compare at least:

- PaddleOCR alone.
- Azure Document Intelligence alone.
- OCR with and without YOLO region detection.
- OCR plus medicine-catalog exact matching.
- OCR plus fuzzy matching.
- OCR plus dosage grammar.
- OCR plus Bengali numeral normalization.
- Full proposed pipeline.

Report field accuracy, complete-row accuracy, review rate, runtime, and false correction introduced by post-processing.

Suggested wording: *The ablation study isolates the contribution of region detection, provider choice, catalog normalization, and dosage-grammar repair to each field-level outcome.*

Completion evidence: A baseline and ablation table supports the main OCR novelty claim.

[] M-C7 Demonstrate transaction-level booking correctness under concurrency

Location: Section 2.2.2 and Listing 1, pp. 6-7; Table 3, p. 12.

Problem: The pseudocode checks slot availability and then writes a new status but does not show row locking, atomic conditional update, serializable isolation, or a uniqueness constraint. A fast real-time update does not prove that double booking is impossible.

Why this matters: Concurrent booking is a correctness problem, not only a latency problem.

Required correction:

- Implement an atomic transaction using a row lock, conditional update, or equivalent database mechanism.
- Add a uniqueness constraint preventing overlapping active appointments.
- Make repeated client requests idempotent.
- Publish the real-time event only after successful commit.
- Test many simultaneous clients attempting the same slot.
- Report the number of successes, conflicts, duplicate bookings, rollbacks, retries, and propagation time after commit.

Suggested wording: *Booking is completed by an atomic database transaction that locks or conditionally updates the target slot under a uniqueness constraint. A real-time event is published only after a successful commit.*

Completion evidence: A reproducible concurrency test shows zero duplicate active appointments under the reported workload.

[] M-C8 Add formal AI-assistant safety and factuality evaluation

Location: Section 2.2.5, Listing 3, pp. 8-9; Section 4, pp. 9-12.

Problem: The assistant is described as safe because of prompts, route restrictions, and schema validation. These controls do not by themselves prevent hallucination, unsafe advice, privacy leakage, or prompt injection.

Why this matters: The assistant can summarize health records and influence navigation, so unsafe or fabricated output is a major rejection and deployment risk.

Required correction: Build a reproducible test suite including:

- Allowed navigation requests and unsupported routes.
- Diagnosis and prescription requests.
- Emergency symptoms and crisis language.
- Unauthorized patient-record requests.
- Missing, expired, or revoked consent.
- Prompt injection and attempts to override system instructions.
- Conflicting records, negation, absent information, and unsupported certainty.
- Provider failure, timeout, malformed schema, and fallback behavior.
- Bengali and English versions of each important category.

Report intent accuracy, navigation success, safe-refusal rate, unsafe-completion rate, schema-validity rate, hallucination rate, source-grounding rate, and PII leakage.

Suggested wording: *The assistant is an interface-support component. It does not diagnose, prescribe, independently modify records, or replace review of the source record. Record summaries are drafts and must be verified by the clinician.*

Completion evidence: A reproducible safety benchmark and failure-case appendix are included.

[] M-C9 Add ethics and data-governance details for prescription images and health records

Location: Entire manuscript; no dedicated ethics/data statement is provided.

Problem: The manuscript mentions an OCR corpus of approximately 100 annotated images but does not state whether the prescriptions are real, how they were obtained, whether identifiers were present, or what approval or consent applied.

Why this matters: Prescriptions and medical histories can contain sensitive patient information, signatures, physician identifiers, and institutional data.

Required correction:

- State whether the corpus is real, synthetic, public, or mixed.
- Identify the approving authority, ethics review, waiver, permission, or consent basis.
- Describe de-identification, annotation access, storage, retention, deletion, and redistribution restrictions.
- Provide a separate data-availability statement and data license.
- Explain whether raw images can be shared; if not, release synthetic or de-identified field crops and evaluation outputs.
- Confirm that screenshots and demo accounts use synthetic records only.

Suggested wording: The OCR evaluation used <describe source> prescription images under <approval or permission>. Direct identifiers and signatures were removed before annotation; raw images are not publicly distributed unless explicit release permission was obtained.

Completion evidence: The ethics, privacy, and data-availability statements are specific and match the supplementary package.

[] M-C10 Remove production-grade and clinically effective wording

Location: Abstract; Sections 1, 4, and 5, pp. 1, 3, and 11-13.

Problem: The system has no reported field pilot, independent security audit, clinical validation, or large representative OCR corpus, yet some wording implies production readiness and operational benefit.

Why this matters: Overclaiming in health software can lead to immediate rejection and unsafe interpretation.

Required correction:

- Describe Medora as a *research and pilot implementation*.
- Replace statements that patients stop re-explaining history or that the app reduces burden with intended or testable benefits.
- State explicitly that clinical effectiveness, security certification, and regulated deployment have not been established.
- Do not claim that schema validation makes an AI model safe.
- Ensure the public deployment carries an appropriate demonstration disclaimer.

Suggested wording: Medora is a feature-complete research and pilot implementation. Its clinical effectiveness, security posture, and operational impact require independent evaluation before real-world healthcare deployment.

Completion evidence: The abstract, impact, conclusion, deployment page, and README use evidence-aligned maturity language.

3. Major revisions - workflow, benchmarks, and reproducibility

[] M-M1 Rewrite the opening motivation in academic style

Location: Section 1, pp. 1-2.

Problem: The opening uses conversational statements about patients carrying information gaps and prescriptions returning as handwriting that the next doctor cannot read.

Why this matters: A journal manuscript should define the workflow problem precisely and support its contextual claims.

Required correction:

- Replace anecdotal phrasing with a structured description of fragmented appointment, record, prescription, and communication processes.
- Cite relevant Bangladesh health-information or digital-health sources.
- Avoid implying that all prescriptions are unreadable or that every patient uses disconnected systems.
- Separate measured context from design assumptions.

Suggested wording: *Healthcare workflows in Bangladesh are frequently distributed across separate appointment, record-management, prescription, and communication processes. This fragmentation can reduce continuity when information generated during one encounter is not readily available during subsequent care.*

Completion evidence: The motivation is precise, cited, and avoids anecdotal generalization.

[] M-M2 Define symptom-to-specialty search as navigation rather than triage

Location: Sections 1 and 2.2.2, pp. 4 and 6.

Problem: Mapping symptoms to a specialty can be interpreted as diagnosis or clinical triage, but no clinical validation, urgency model, or safe uncertainty handling is reported.

Why this matters: Incorrect routing could delay care, particularly for urgent symptoms.

Required correction:

- State that the component suggests candidate specialties for navigation only.
- Implement deterministic emergency guidance outside the language model.
- Include an *uncertain or unable to classify* output.
- Allow the user to correct or bypass the suggestion.
- Evaluate against clinician-labeled examples and report agreement and unsafe-routing cases.
- Do not assign urgency unless separately validated.

Suggested wording: *The search component suggests candidate specialties for navigation and does not diagnose a condition, determine urgency, or replace emergency-care guidance.*

Completion evidence: The interface, manuscript, and evaluation use navigation terminology and provide safe uncertainty handling.

[] M-M3 Ground clinician record summaries in source data

Location: Sections 2.2.5 and 3, pp. 8-9.

Problem: The assistant can summarize a patient record, but the manuscript does not explain how factuality, omissions, negation, conflicting entries, temporality, and unsupported inference are controlled.

Why this matters: A misleading summary can cause clinical error even when it is schema-valid.

Required correction:

- Generate summaries from structured fields wherever possible.
- Link every summary item to its source record or document section.
- Use *not found in the record* rather than allowing the model to fill missing information.
- Test allergies, dosage, negation, temporal sequence, duplicates, and conflicting entries.
- Prevent automatic write-back to the medical record.
- Display a mandatory clinician-verification banner.

Suggested interface text: *AI-generated draft summary. Verify every item against the linked source record before clinical use.*

Completion evidence: Factuality and source-grounding results are reported and summaries remain advisory.

[] M-M4 Correct the meaning of patient acceptance or rejection of a prescription

Location: Section 2.2.3, p. 7.

Problem: The paper states that a patient can accept or reject a prescription and that this keeps the record honest about what was filled. A patient action does not verify dispensing or adherence.

Required correction:

- Rename the action *receipt acknowledgment* or *report discrepancy*.
- Separate prescription issuance, patient acknowledgment, pharmacy dispensing, and medication adherence.
- Do not infer compliance or clinical approval from the button.
- Update database fields and interface labels if the implementation currently uses misleading semantics.

Suggested wording: *The patient may acknowledge receipt or report a discrepancy in the displayed prescription. This action does not represent dispensing confirmation, clinical approval, or medication adherence.*

Completion evidence: Database, interface, API, and manuscript use the corrected event semantics.

[] M-M5 Provide a fully reproducible benchmark protocol

Location: Section 4 and Table 3, pp. 11-12.

Problem: Table 3 reports representative latency and web-vital values without the number of runs, concurrency, hardware, region, payload, cache state, data volume, or test date.

Why this matters: Representative numbers are not reproducible evidence.

Required correction:

- Report request count, warm-up procedure, concurrency, endpoint mix, payload sizes, database size, hardware, software versions, cloud region, and network path.
- Separate local application latency from external provider latency.
- Repeat runs and report median, p95, p99, standard deviation or range.
- Archive Locust, k6, and Lighthouse configurations and raw outputs.
- Explain how target values were selected and whether they are engineering service-level objectives.
- Record the exact release and date of each benchmark.

Suggested wording: Performance values are reported for a fixed archived release under the documented hardware, region, workload, and cache conditions. Raw benchmark output and scripts are included with the release.

Completion evidence: All values in the revised table can be reproduced using documented commands.

[] M-M6 Separate real-time propagation latency from data consistency

Location: Sections 2.2.2 and 4; Table 3, pp. 6 and 11-12.

Problem: A roughly 500 ms UI update is presented as keeping slots consistent, but propagation speed does not prove correctness during disconnection, retries, event loss, or stale client state.

Required correction:

- Measure from database commit to visible client update.
- Test multiple clients, weak networks, reconnection, missed events, duplicate events, and out-of-order events.
- Require authoritative slot revalidation before final booking.
- Provide a refresh or re-query mechanism if the real-time event is missed.
- Report database correctness and UI freshness as separate outcomes.

Suggested wording: The database transaction is the source of truth; the real-time channel provides a best-effort UI update, and clients revalidate slot state before booking.

Completion evidence: Consistency and propagation are evaluated as separate properties.

[] M-M7 Document offline caching of sensitive data

Location: Section 2.2.6, p. 8.

Problem: The paper states that the PWA caches offline but does not identify whether protected records, prescriptions, API responses, or tokens can persist on the device.

Why this matters: Offline caching can expose health information on shared, lost, or compromised devices.

Required correction:

- List exactly which resources are cached by the service worker.
- Exclude protected API responses and prescription images from persistent cache unless a secure design is documented.
- Define token storage, expiration, refresh, and logout cleanup.
- Restrict background sync from transmitting or persisting sensitive payloads unexpectedly.
- Add tests that inspect cache contents before and after login/logout.
- State the limitations of offline use clearly to users.

Suggested wording: *Offline caching is restricted to static application assets and non-sensitive navigation resources; protected medical records and document images are excluded from persistent service-worker caches.*

Completion evidence: Use this wording only after the implementation and automated tests confirm it.

[] M-M8 Add a detailed role-permission and record-access matrix

Location: Sections 2.1, 2.2.1, 2.2.5, and 2.2.6, pp. 5-8.

Problem: Role-based access is described generally, but the exact permissions for patients, doctors, administrators, OCR, summaries, exports, audit logs, and record edits are unclear.

Why this matters: Health-data access depends on role, resource ownership, patient-clinician relationship, and consent scope.

Required correction: Add a matrix covering whether each role can:

- View and edit profile/medical history.
- View another patient's data.
- Upload or download prescriptions and reports.
- Run OCR or request AI summaries.
- Create or revise a prescription.
- Export records.
- Verify doctors.
- View audit logs.
- Revoke consent or delete data.

Test every denied path at API level; frontend route guards must not be treated as the security boundary.

Suggested wording: *Authorization is enforced on the backend using role, patient-clinician relationship, consent scope, resource ownership, and appointment/consultation state. Frontend route guards are usability controls only.*

Completion evidence: The matrix and automated negative tests cover every protected action.

[] M-M9 Specify consent semantics and revocation behavior

Location: Sections 2.1 and 2.2.6; Figure 6(d), pp. 5 and 8-11.

Problem: Fine-grained consent is mentioned without a precise default state, data scope, recipient, purpose, duration, revocation effect, historical access, external AI processing, or research use.

Why this matters: Patient-controlled sharing is a central contribution and must be technically unambiguous.

Required correction:

- State whether the default is deny or allow.
- Define record categories, recipient, purpose, validity period, and revocation.
- Explain whether a clinician retains previously accessed data after revocation.
- Separate clinical-sharing consent, external AI-processing consent, and research/data-export consent.
- Explain what cannot be retracted after lawful disclosure to a provider.
- Ensure the patient-visible audit trail records meaningful access events.

Suggested wording: *Consent records are versioned by scope, recipient, purpose, and validity period. Revocation blocks future access but cannot retract data already lawfully disclosed to an external service or clinician.*

Completion evidence: Consent rules are represented consistently in the data model, interface, tests, privacy documentation, and manuscript.

[] M-M10 Expand and balance the related-work comparison

Location: End of Section 1, p. 4.

Problem: OpenMRS and OpenEMR are described as heavy and not designed around Medora-like flows, but no systematic comparison is supplied.

Why this matters: The article should explain scope differences without dismissing mature clinical information systems.

Required correction: Add a table comparing:

- Deployment model and infrastructure requirements.
- Patient-facing mobile/PWA workflow.
- Appointment booking and real-time slot updates.
- Prescription management and image OCR.
- Consent and audit mechanisms.
- Bengali localization.
- AI navigation or summarization.
- Interoperability and data standards.
- Open-source status and validation evidence.

Suggested wording: *Institutional electronic-record platforms address broader clinical and administrative requirements, whereas Medora focuses on the integrated mobile-first workflow evaluated in this study.*

Completion evidence: The novelty statement is balanced and based on explicit comparison criteria.

[] M-M11 Document provider and model configuration for reproducibility

Location: Code metadata; Section 2.1; Figure 2, pp. 2 and 5-6.

Problem: External LLM and OCR providers can change, but model identifiers, API versions, decoding settings, fallback order, timeout, retention settings, and access dates are not fully documented.

Why this matters: Provider drift can change behavior and make results impossible to reproduce.

Required correction:

- Report provider, model ID, API version, date, region, temperature/decoding, timeout, retry policy, fallback order, schema, context limit, and retention setting.
- Report provider-specific results instead of silently pooling them.
- Provide a deterministic mock provider for tests and demonstrations.
- Archive test prompts, expected schema outputs, and fixed provider-response fixtures where licensing permits.
- State which functions require paid or cloud services.

Suggested wording: External services are configuration-dependent and may change. The archived release records the exact provider/model versions and includes a deterministic mock mode for reproducibility.

Completion evidence: Core tests and demonstrations can run without paid external accounts.

[] M-M12 Add a complete illustrative example with safety checkpoints

Location: Section 3, pp. 8-9.

Problem: The current examples are narrative and do not provide inputs, intermediate outputs, confidence values, consent state, or reproducible fixtures.

Why this matters: The example should demonstrate the distinctive OCR and AI-boundary contributions, including their safeguards and failure behavior.

Required correction: Provide two compact reproducible examples:

1. **Prescription example:** synthetic image, detected region, raw OCR, catalog match, dosage repair, confidence, review flag, human correction, and final structured record.
2. **Assistant example:** role check, relationship check, consent check, redacted prompt, schema output, navigation, and a safe refusal for a prohibited request.

Include the exact fixtures, expected outputs, and commands in the archive.

Suggested wording: All example data are synthetic. The complete fixtures and expected outputs are included in the `examples` directory of the archived release.

Completion evidence: The examples run without private data and do not imply clinical decision-making.

4. Presentation, figures, and language

[] M-P1 Revise the title and abstract to remove absolute privacy language

Location: Title and Abstract, p. 1.

Problem: The abstract contains precise accuracy, speed, and anonymity claims without enough evaluation context and limitations.

Required correction:

- Use the recommended title or an equivalent focused title.
- State the OCR evaluation scope and review-gated nature of extracted fields.
- Describe consent and redaction as controls that reduce exposure, not as anonymity guarantees.
- Identify the artifact as a research and pilot platform.
- Avoid reporting approximate benchmark values in the abstract unless denominators and evaluation conditions are available in the main text.

Guidance: Use the suggested abstract at the beginning of this report after inserting verified dataset and release details.

Completion evidence: The abstract includes no guarantee of anonymity, clinical safety, or production readiness.

[] M-P2 Redesign Figures 1-3 around trust boundaries and data types

Location: Figures 1-3, pp. 5, 6, and 9.

Problem: The current diagrams do not clearly distinguish de-identified text, raw images, audio, local services, external providers, data stores, and audit records.

Required correction:

- Draw a clear local trust boundary and place external providers outside it.
- Use distinct arrow labels for text, image, audio, metadata, and real-time events.
- Mark optional versus required cloud components.
- Show where direct identifiers may exist, where redaction occurs, and what is logged.
- Show the OCR image path separately from the text-model path.
- Increase font size and remove decorative explanatory text from the diagram.

Suggested caption guidance: State explicitly which data may leave the deployment and under which configuration.

Completion evidence: A reader can trace every sensitive-data path from input to storage or provider.

[] M-P3 Reduce manuscript length and remove repeated AI-boundary explanations

Location: Sections 1, 2.1, 4, and 5.

Problem: The same claims about one AI boundary, anonymization, schema validation, and provider rotation are repeated several times.

Required correction:

- Explain the architecture once in Section 2.1.
- Use the Impact section for demonstrated technical/research utility rather than another architecture summary.
- Move long pseudocode, route counts, and provider implementation detail to developer documentation.
- Retain only algorithms that explain distinctive correctness or safety mechanisms.
- Use the saved space for evaluation, limitations, ethics, and reproducibility.

Guidance: Each major feature should be introduced once and supported by a result, test, or worked example.

Completion evidence: The manuscript is substantially shorter without loss of essential evidence.

[] M-P4 Correct figure captions and terminology

Location: Figures 4-6, pp. 10-11.

Problem: Some captions do not precisely match the displayed screens, and terms such as Bangla/Bengali, doctor/clinician, accept/reject, and prescription view/OCR review are inconsistent.

Required correction:

- Verify every panel against the archived release.
- Use *Bengali* consistently in academic prose.
- Use *clinician* where the role is not limited to a physician.
- Correct prescription acknowledgment terminology.
- Label all screenshots as synthetic demonstration data.
- Increase panel size and crop unnecessary blank space.

Suggested caption: Figure 4(d): Prescription-upload and OCR-review interface showing extracted text and fields requiring user verification.

Completion evidence: Every caption accurately describes the visible panel and revised workflow.

[] M-P5 Verify contextual statistics and avoid causal linkage

Location: Section 1, p. 2.

Problem: The physician-density value is used to infer that doctors cannot spend time reconstructing patient history, but the data year and indicator definition are not given and the causal link is unsupported.

Required correction:

- Report the exact data year, indicator definition, source, and access date.
- Remove the direct inference about consultation time unless supported by a specific study.
- Avoid broad claims that most users rely on mid-range Android phones or patchy data without evidence.
- Separate the design target from a measured population characteristic.

Suggested wording: *Bangladesh reports a comparatively low physician density in the cited data source. This context motivates efficient information access, although the present study does not measure consultation duration or workforce burden.*

Completion evidence: Every contextual statistic is dated, defined, and used only for a supported inference.

[] M-P6 Revise the conclusion to foreground current limitations

Location: Section 5, pp. 12-13.

Problem: The conclusion repeats strong privacy and safety claims but gives less prominence to absent clinical validation, independent security assessment, and field evaluation.

Suggested conclusion: *Medora integrates healthcare workflow management, prescription digitization, consent-controlled information sharing, and bounded AI assistance in an open-source mobile-first platform. Its technical evaluation demonstrates feasibility under the reported test conditions, but the system has not undergone clinical validation, independent security assessment, or field-effectiveness evaluation. Future work will prioritize a larger prescription corpus, formal privacy and security testing, grounded summarization evaluation, and prospective usability assessment.*

Completion evidence: The conclusion no longer guarantees anonymity, safety, or production readiness.

Final pre-submission audit

Privacy and safety

- [] The external text, image, and audio data paths are documented separately and accurately.
- [] No anonymity, non-traceability, or zero-leakage guarantee remains.
- [] PII redaction, consent enforcement, unauthorized access, prompt injection, safe refusal, and factuality are evaluated.
- [] All AI-generated or OCR-extracted clinical content is clearly review-gated.

Core correctness and evaluation

- [] Atomic booking correctness is demonstrated under concurrent requests with zero duplicate active appointments.
- [] OCR metrics include transparent units, confidence intervals, review coverage, baselines, and ablations.
- [] Performance metrics include complete workload, environment, repetitions, raw outputs, and reproducible commands.
- [] Real-time propagation, database consistency, reconnection, and stale-client behavior are evaluated separately.

Ethics and governance

- [] Prescription-image provenance, approval or permission, de-identification, storage, and release restrictions are documented.
- [] Role, relationship, consent, revocation, export, audit, cache, and deletion semantics are defined and tested.
- [] The manuscript states that independent security assessment and clinical validation have not been completed.

Release and presentation

- [] A fixed release, archive DOI, commit hash, citation file, changelog, and reproducibility package are available.
- [] Figures show trust boundaries and sensitive-data flows accurately and are readable at journal scale.
- [] Title, abstract, impact, and conclusion use research/pilot language and distinguish demonstrated from prospective benefits.
- [] All references, URLs, figure numbers, metadata fields, and declarations have been checked.

Required response from the student

Submit all of the following together:

1. Revised manuscript with tracked changes or a detailed change log.
2. Point-by-point response matching every checklist code in this report.
3. Fixed software release link, archive DOI, and exact commit hash.
4. Reproducibility package containing tests, benchmark scripts, configuration, synthetic/shareable fixtures, and raw outputs.